Obfuscation in a Heterogeneous Pricing Market: When Does It Survive a Learning Producer?

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Abstract

We extend a single-consumer Bayesian pricing game to a heterogeneous market in which consumers have private willingness-to-pay and each chooses its own obfuscation strategy against a learning producer. The producer makes little parametric assumption about the demand distribution; we show its optimal pricing reduces to a two-dimensional sufficient statistic (price and local hazard rate), making the problem tractable for deep reinforcement learning. In a two-sided multi-agent DQN setting, obfuscation survives against a best-responding, belief-updating producer ($\bar{\epsilon}^* < 1$) and increases significantly with the consumer discount factor β ($\rho = -0.77$, $p < 10^{-6}$), saturating at high patience. Comparing individual and cooperative consumers reveals obfuscation as a negative externality of decentralised choice. An ablation against a fixed-price (non-learning) producer shows obfuscation falls toward truthful play, confirming the producer's learning amplifies it.

1 Introduction

We consider a monopolistic producer. A monopolist restricted to a single price leaves consumers with positive surplus, but one that can infer individual willingness-to-pay can price-discriminate — in the limit of perfect discrimination, extracting all consumer surplus. Since such extraction is enabled by information, consumers have an incentive to obfuscate: by corrupting the producer's inference of their willingness-to-pay, they defend the surplus that perfect discrimination would erase. This motivates our central question: does obfuscation survive when the monopolist actively learns?

In our previous paper, we built a simple Bayesian game with a single consumer. This paper extends that setting to a small niche market with heterogeneous consumers each with a different willingness to pay. The producer does not make any parametric assumption about the distribution of consumers' willingness to pay (except regularity conditions, e.g. an atomless F with a density), but only seeks the optimal quantile that maximise the profits. This yields a general Bayesian game well-suited to a DQN. While a fully non-parametric approach can be intractable when maintaining the entire posterior over distribution, we argue that two statistics - the optimal price and the local hazard rate - characterise the producer's learning, making the model tractable. As before, obfuscation introduces strategic noise in demand realisation that attenuates the mapping from willingness-to-pay to observed acceptance rates, thereby distorting the producer's inference of the hazard rate.

2 Related Work

Our setting is closest to Amin, Rostamizadeh, and Syed[1, 2] who show analytically that a single strategic buyer can hide her value from a learning seller, and that the seller's advantage depends on the buyer discounting future surplus. We arrive at a related conclusion through a different route — reinforcement learning in a two-sided game with heterogeneous buyers — and additionally characterise the obfuscation–patience relationship $\bar{\epsilon}^*(\beta)$ empirically, including its saturation at high β. Unlike the no-regret seller of [1, 2] our producer is belief-greedy (myopic); a fully strategic producer is left to future work.

2.1 Obfuscation of Information

Rhodes and Zhou [7] study a model in which consumers choose whether to share their information with a price-personalising firm, a choice that affects their utility. Under certain regularity conditions, the privacy choice admits an interior Nash equilibrium that deviates from the socially optimal level. Notably, they find that individual privacy choices impose negative externalities on other consumers - echoing the externality we identify in our obfuscation setting. Their work provides a theoretical basis for our model of consumers' obfuscation.

2.2 Algorithmic Collusion

Calvano et al. [5] use tabular Q-learning to demonstrate that collusion can emerge even in the simple algorithmic setting. Their tabular Q-learning setting is close in spirit to our single-consumer model: the state is the recent price history (bounded memory), rendering the repeated pricing game Markov, and agents learn profit-maximising strategies from experience rather than solving the model analytically. Strikingly, independent learners autonomously converge to supra-competitive collusive prices, sustained by self-learned reward-punishment strategies - without being explicitly designed to collude. Using DQN, we further show a discrepancy between the aggregate (cooperative) and individual decision settings — a gap we characterise as a negative externality of decentralised choice.

2.3 Equilibrium Learning in Dynamic Pricing

Liu [6] presents Nash-equilibrium learning for n-player dynamic pricing using a deep policy-network. The pricing game is modelled as a Markov game equipped with a best-response oracle (a function returning each player's optimal reaction), which lets the joint n-player strategy optimisation be reframed as minimising a deviation gap δ - the unilateral incentive to deviate, which vanishes at a Nash equilibrium. Minimising δ yields convergence of the learning dynamics to the approximate Nash equilibrium. In contrast, our producer learns a value-based best response online - without access to a best-response oracle - and consumers strategically obfuscate to corrupt that learned response. Equilibrium thus emerges from two-sided, adversarial learning rather than oracle-driven policy optimisation.

3 Theoretical Base

In this game, we have N heterogeneous consumers, each with private willingness to pay W_i. Heterogeneity enters only through W_i; obfuscation ϵ_i is a strategic choice separate from the type W_i, though its optimal value may depend on W_i. Each consumer's policy is mixed :

$$\epsilon_i \mathbb{1}(W_i \geq x_t) + \frac{(1-\epsilon_i)}{2}, \ \ \epsilon_i \in [0,1]$$

where ϵ_i controls the level of obfuscation. Consumers try to learn optimal ϵ_i^* via DQN.

The producer is single and follows a belief-greedy strategy: it maintains an estimate $\hat{A}_t(\cdot)$ of the acceptance curve $\tilde{S}(\cdot;\bar{\epsilon})$ updated online from observed accept-rates, and prices greedily: $x_t = \arg\max_x (x-c)\hat{A}_t(x)$. By the theorem in section 2.1, the producer's optimal price depends on $\hat{A}_t$ only through the two local statistics $(x_t, \tilde{h}_t)$ - the current price and the local effective hazard- which we therefore take as its state. The motivation for this dimension reduction is statistical : with N consumers the empirical CDF satisfies $\sup_x |\hat{F}_N(x) - F(x)| = O_p(N^{-1/2})$ [3, 4], so under the atomless assumption, the plug-in estimates $(\hat{x}_t, \hat{\tilde{h}}_t)$ are consistent, so a market with sufficiently large N lets the producer recover the two-dimensional statistic with high probability. The producer's expected profit at price x, given aggregate obfuscation $\bar{\epsilon}$, is

$$\Pi(x;\bar{\epsilon}) = (x-c)\tilde{S}(x;\bar{\epsilon}), \qquad \tilde{S}(x;\bar{\epsilon}) := \bar{\epsilon}(1-F(x)) + \tfrac{1-\bar{\epsilon}}{2}$$

where c is the unit production cost, $\tilde{S}$ is the effective demand. The constant term $\frac{1-\bar{\epsilon}}{2}$ is a price insensitive noise floor of acceptances produced by randomisation.

3.1 Reduction of Dimensionality

Lemma 3.1. *Assume F is differentiable with density f and $\tilde{f}(x) = \bar{\epsilon}f(x) > 0$ at any interior maximiser. Then every interior optimal price x^* satisfies*

$$x^* = c + \frac{\tilde{S}(x^*;\bar{\epsilon})}{\tilde{f}(x^*)} = c + \frac{1}{\tilde{h}(x^*;\bar{\epsilon})}$$

pf. $\Pi(x;\bar{\epsilon}) = (x-c)\tilde{S}(x;\bar{\epsilon})$ is differentiable with $\Pi'(x) = \tilde{S}(x) - (x-c)\tilde{f}(x)$. At an interior maximiser $\Pi'(x^*) = 0$; dividing by $\tilde{f}(x^*) > 0$ gives $x^* - c = \tilde{S}(x^*)/\tilde{f}(x^*) = 1/\tilde{h}(x^*)$. Since $\tilde{S}(x;\bar{\epsilon}) \to \frac{1-\bar{\epsilon}}{2} > 0$ as x grows, the noise floor can push the maximiser to the upper boundary of the price domain; an interior optimum need not exist for $\bar{\epsilon} < 1$. This distortion relative to the truthful benchmark ($\bar{\epsilon} = 1$) is the mechanism by which obfuscation operates. We therefore work on a bounded price domain $x \in [0, \bar{x}]$ on which the lemma characterises any interior optimum and Kakutani gives existence.

Theorem 3.1 (Sufficient Statistic)**.** *The producer's local optimal pricing depends on F only through the local hazard function near x^*.*

pf. By the previous lemma, any interior optimum solves $x = c + 1/\tilde{h}(x;\bar{\epsilon})$, which involves F only through $\tilde{h}(x;\bar{\epsilon}) = \bar{\epsilon}f(x)/\tilde{S}(x;\bar{\epsilon})$ evaluated near x. Hence two type distributions inducing the same effective hazard on a neighbourhood of their optimum share the same interior optimal price; the optimality check depends on F only through the local pair $(x, \tilde{h}(x;\bar{\epsilon}))$.
$\rightarrow$ the producer's learning state can be taken to be the two-dimensional $(x_t, \tilde{h}_t)$ which we use as the input to the DQN, paralleling the (μ, σ) state of the single-type model.

3.2 Existence of Equilibrium in the Game

Assume F, the CDF of consumer willingness-to-pay, is atomless. Then the expected demand $\tilde{S}$ is a continuous function of $(x, \bar{\epsilon})$ even though individual accept

indicators jump - the discontinuity gets integrated out over the type distribution F. By Berge's maximum theorem, the producer's best-response correspondence $X^*(\bar{\epsilon}) = \arg\max_x \Pi(x;\bar{\epsilon})$ is nonempty-valued and u.h.c.; the induced aggregate best response map on $\bar{\epsilon} \in [0,1]$ is u.h.c. with nonempty, convex (from mixed obfuscation strategies), compact values. Therefore by the Kakutani's fixed point theorem, the fixed point (Bayes-Nash equilibrium) exists.
→ DQN dynamics approximate this adjustment; the experiment tests whether they converge to the fixed point promised.

4 Empirical Results

The producer observes the consumers' actions, updates its belief about the acceptance curve, and reprices accordingly. Each consumer maximises its own discounted surplus, knowing producer adapts its price to aggregate behaviour. In this game, we vary β, the time discounting factor, to test whether patience correlates with obfuscation. As β increases, the equilibrium $\bar{\epsilon}^*$ decreases - more

Table 1: Equilibrium aggregate obfuscation $\bar{\epsilon}^*$ in the dynamic game with a learning producer, as a function of the consumer discount factor β (3 seeds each, $T = 25000$). For all $\beta > 0$, $\bar{\epsilon}^* < 1$: obfuscation survives against a best-responding, belief-updating producer, and $\bar{\epsilon}^*$ decreases with patience (i.e. obfuscation increases).

β	$\bar{\epsilon}^*$ (mean ± std)
0.00	0.936 ± 0.024
0.50	0.828 ± 0.042
0.90	0.689 ± 0.077
0.99	0.717 ± 0.054

patient consumers obfuscate more, since discounted future gain from corrupting the producer's belief outweighs the immediate cost. The trend is monotone up to $\beta = 0.90$; the slight uptick at $\beta = 0.99$ lies within seed variation. Thus we reran the experiment with different seed settings.

With 24 seeds, $\beta = 0.99$ is statistically indistinguishable from $\beta = 0.9$: the trend decreases and then plateaus rather than reversing. The overall negative correlation is highly significant (Spearman $\rho = -0.769$ with $p < 10^{-6}$). The plateau suggests the obfuscation incentive saturates once consumers are sufficiently patient.

Table 2: Equilibrium aggregate obfuscation $\bar{\epsilon}^*$ vs. consumer discount factor β (24 seeds, $T = 20000$). For all $\beta > 0$, $\bar{\epsilon}^* < 1$: obfuscation survives a best-responding, belief-updating producer, and $\bar{\epsilon}^*$ decreases with patience (i.e. obfuscation increases), saturating near $\beta \approx 0.9$. Spearman over the raw $(\beta, \bar{\epsilon}^*)$ pairs ($n = 96$).

β	$\bar{\epsilon}^*$ (mean $\pm$ std)
0.00	0.914 ± 0.035
0.50	0.852 ± 0.048
0.90	0.740 ± 0.071
0.99	0.749 ± 0.059
Spearman $\rho = -0.769$, $p < 10^{-6}$	

4.1 Convergence

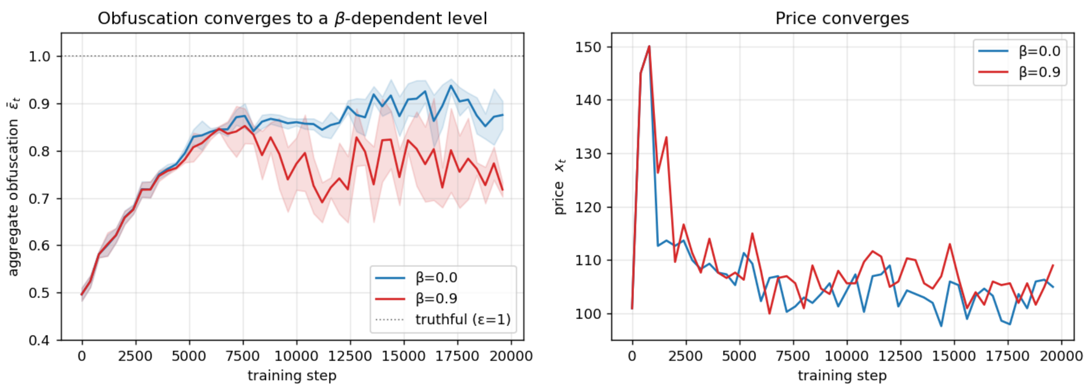

Figure 1: Convergence of the two-sided learning dynamics (mean over 3 seeds, shaded ± 1 std). *Left:* aggregate obfuscation $\bar{\epsilon}_t$ settles to a β-dependent level — higher for myopic consumers ($\beta = 0$, near the truthful benchmark) and lower for patient ones ($\beta = 0.9$). The residual gap from $\bar{\epsilon} = 1$ at $\beta = 0$ reflects estimation noise on a value surface that is flat in ϵ. *Right:* the producer's price x_t converges to the profit-maximising level, independent of β. The dynamics settle rather than cycle, supporting the existence claim of Section 2.2.

We track the trajectories of aggregate $\bar{\epsilon}$ for two β values; they settle near 0.91 ($\beta = 0$) and near 0.74 ($\beta = 0.9$) respectively. The $\beta = 0.9$ run oscillates more than $\beta = 0.0$, so the convergence is to a band rather than a razor-sharp point. The price, it spikes during early exploration stage, then drops and converges to 105 by step 7500 and remains there.

4.2 Externalities

Obfuscation carries a private cost - randomising one's response forfeits surplus on the current offer - but a shared benefit: it corrupts the producer's belief and lowers the price for all consumers in future rounds. The strategic consequence of this cost/benefit split is not obvious a priori, so we compare two regimes in the dynamic game: a cooperative consumer that chooses $\bar{\epsilon}$ to maximise aggregate discounted surplus and individual consumers each maximising their own discounted surplus. The result shows that the cooperative consumer chooses near-truthful

Table 3: Equilibrium $\bar{\epsilon}^*$ under cooperative (aggregate-surplus-maximising) vs. individual consumers, dynamic game, 8 seeds. Obfuscation is sustained only under individual optimisation; the gap widens with β.

β	individual	cooperative	gap
0.00	0.893 ± 0.022	0.993 ± 0.007	0.099
0.50	0.833 ± 0.039	0.993 ± 0.007	0.160
0.90	0.750 ± 0.068	0.980 ± 0.024	0.230
0.99	0.755 ± 0.056	0.979 ± 0.026	0.224

play, whereas individual consumers obfuscate substantially more. Contrary to the simple free-rider intuition (under which individuals would obfuscate less), obfuscation here is sustained only under decentralised, individual optimisation; the gap in the table above, which widens with β, is another form of externality - each consumer's obfuscation degrades the producer's information in a way the consumers would collectively prefer to limit but individually cannot resist. This may explain why the aggregate model collapses to truthful play while the individual model does not.

5 Ablation

To test whether obfuscation is genuinely driven by the producer's learning, we compare against a heuristic baseline : a no-learning producer that charges a fixed price and never updates. Because the price is constant, consumers have no belief to corrupt, so the intertemporal channel through which obfuscation operates is largely removed (though a within-period effect remains, as we discuss below). Under the fixed-price seller, the equilibrium obfuscation rate increases relative to the learning producer (consumers stay more truthful to their true preferences), but does not reach 1. This is likely because each consumer's own randomisation still affects its own surplus even when the price is fixed. Learning is therefore a driver of obfuscation, but not the sole driver.

Table 4: Ablation: equilibrium obfuscation $\bar{\epsilon}^*$ under a learning (belief-updating) vs. fixed-price (no-learning) producer, dynamic game, 8 seeds. Obfuscation is uniformly higher (less truthful) under the learning producer, confirming that belief-updating amplifies the obfuscation incentive.

β	learning seller	fixed-price seller
0.00	0.893 ± 0.022	0.958 ± 0.018
0.50	0.833 ± 0.039	0.922 ± 0.022
0.90	0.750 ± 0.068	0.817 ± 0.037
0.99	0.755 ± 0.056	0.759 ± 0.102

6 Conclusion

In a real economy, consumer and producer play a stage game; producer sets price, the consumer strategically decides whether to take it or not, and then after observing consumer's movement, the producer adjusts its price strategically and repeats this. The game they play is the traditional Bayesian game, both maximising their own objective function. The result is well-aligned with the one in the previous paper with simpler setting, $\epsilon^* < 1$ and statistically significant obfuscation rising trend in the time discounting factor. Crucially, this survival is intertemporal: with myopic consumers ($\beta = 0$) obfuscation unravels to truthful play, and only the discounted future gain from corrupting the producer's belief sustains it. The simulation shows regardless of an individual's type (willingness-to-pay), agents can rationally obfuscate to improve their objective function. This is actually a negative externality of decentralised choice, given the result that they collectively would prefer to limit it. We established the existence via Kakutani; local convergence is observed empirically with a formal contraction result left to future work.

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